Aggregate / Group Functions:

- Group Functions, as the name suggests, are functions that operate on groups(sets) of values and returns one result per group.
- Group function returns a single result row for every group of queried rows.
- Based on the query statement it may return single or multiple rows.
- The functions that are used as Aggregate Functions are:

    - SUM
    - AVG
    - MAX
    - MIN
    - COUNT

Consider the following table: Employee

AVG: Returns an average value, ignoring null values.

Syntax:
AVG([DISTINCT] column_name)

Ex: SELECT AVG(salary) as “Average Salary” FROM Employee;

Output :

Average Salary

21500
The above query displays the average salary of all the employees in the table Employee

MAX: Returns the maximum value, ignoring null values.

Syntax: MAX([DISTINCT] column_name)

Ex: SELECT MAX(salary) as “Maximum Salary” FROM Employee where Dep_Name='HR';
Output :

Maximum Salary

16000

The above query displays the maximum salary of all the employees in HR Department in the table Employee.

MIN: Returns the minimum value, ignoring null values.

Syntax:
MIN([DISTINCT] column_name)

Ex: SELECT MIN(salary) as “Minimum Salary” FROM Employee where Dep_Name='HR';
Output :

Minimum Salary

10000

The above query displays the minimum salary of all the employees in HR Department in the table Employee.

COUNT: Returns the count of not null values ignoring null values.

Syntax:
COUNT([DISTINCT] column_name)

Ex: SELECT COUNT(DISTINCT Dep_name) Departments FROM Employee;
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The above query displays the count of different departments in the table Employee.

COUNT:(*) Count function with asterisk returns the count of total number of rows including null values

Syntax:
COUNT(*)
Ex: SELECT COUNT(*) FROM Employee;
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The above query displays the total number of rows in table Employee.

GROUP BY clause
Creates a data set, containing several sets of records grouped together based on a condition.

Syntax:
SELECT [,], AGGREGATE FUNCTION() FROM Table_Name GROUP BY [,] ;

Ex: SELECT dep_name,COUNT(emp_id) "No of Employee" FROM Employee GROUP BY dep_name;

Output :

No of Employee
HR 2
Marketing 2
Admin 1

The above query displays the number of employee in each department.

WHERE clause
Used to apply a filter condition before the Grouping the rows.

Syntax:
SELECT [,], AGGREGATE FUNCTION() FROM Table_Name WHERE GROUP BY [,] ;

Ex: SELECT Dep_Name,COUNT(Salary) FROM Employee WHERE Salary>15000 GROUP BY Dep_Name;

Output :

HR 1
Marketing 1
Admin 1
The above query displays department wise count of salary more than 15000.

HAVING clause
Used to apply a filter condition on Aggregated values.
Syntax:
SELECT [,], AGGREGATE FUNCTION() FROM Table_Name WHERE GROUP BY [,] HAVING ;

Ex: SELECT Dep_Name, SUM(Salary) FROM Employee WHERE Salary>12000 GROUP BY Dep_Name HAVING SUM(Salary)<30000;

Output :
HR 16000
Marketing 20000
The above query displays the departments for which total salary is less 30000 excluding the Admin department, total salary for which is 40000.